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Effects of Modified Inclusions and Precipitates Alloyed by Rare Earth Element on Corrosion and Impact Properties in Low Alloy Steel

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Abstract
This study has focused on the morphology and distribution of inclusions and precipitates modified by rare earth (RE) elements, which has a decisive influence on microstructure, corrosion properties and impact behaviors in Q355 low alloy steel. Characterized by the method of electrolytic extraction and ASPEX-scanning electron microscopy (ASPEX-SEM), small-sized spherical RE inclusions have been modified to replace elongated MnS and large-sized Al_2O_3. Q355RE steel after RE alloying has lower corrosion rate and higher value of α/γ, due to the formation of stable and dense rust layers. Q355RE steel also exhibits better resistance to fracture at low temperature, owing to the presence of RE modification to inclusions and its effects on reducing crack initiation and propagation. Nano-scale RE precipitates containing sulfur and phosphorus is observed along grain boundaries by transmission electron microscopy (TEM). The purification of grain boundaries by RE is beneficial to the improvement of corrosion and impact properties.

Keywords Rare earth element · Inclusion · Precipitate · Q355 low alloy steel · Corrosion resistance · Impact toughness

1 Introduction

Steel for civil construction is a mainstream trend of steel usage in China. Compared with traditional brick-wood and concrete structures, steel structure possesses light weight and low energy consumption, and more than 60% of steel materials can be recycled after demolition [1–3]. Q355 low alloy steel has been one of the most widely used building materials in steel structure for a long time [4–6]. However, Q355 steel is usually subjected to atmospheric corrosion related to various factors such as air temperature, humidity, sunshine and pollutant, which may lead to corrosion structure failure [7, 8]. Two common methods are used to alleviate destructive corrosion including multi-alloying and coating. From the perspective of economy and environmental protection, a cheaper and environment-friendly technology is urgent to be developed. Another problem that all structural steels may face is low temperature structure failure. When the ambient temperature is declined to a lower temperature, the brittleness of Q355 steel increases significantly, which probably leads to cold-brittle failure [9–11]. Impact toughness test is an effective method to evaluate the ability of material to resist external impact.

Is there any low-cost way to solve the two problems discussed above? Researchers have indicated that large-sized inclusions are the main causes to corrosion and impact failure [12, 13]. Large-sized inclusions such as Al_2O_3 and MnS are easy to be formed in structural steels during solidification due to the segregation of chemical compositions. It is well-known that large-sized inclusions are hard to be eliminated and always provide initiation sites for pitting corrosion and cracks [14–16]. Some researchers have focused on the advantageous roles of rare earth (RE) elements adding into structural steels. La and Ce are the left-over elements in the extraction of RE elements in China and are not expensive. After adding Ce and La into steels, the morphology and size of inclusions have been changed significantly and stable precipitates containing RE including oxides, sulfides and oxysulfides tend to be formed more readily [17–20].

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Furthermore, nano-scale precipitates containing RE may also be formed, although it is rarely reported. In general, previous research has proved beneficial effects of RE in increasing impact toughness and corrosion resistance properties, while the correlation between inclusions and precipitates and properties has not been established clearly.

Therefore, the purpose of this study is to further investigate the influence of RE on corrosion resistance and impact toughness on the basis of modifying inclusions and precipitates. The comparison between common Q355 steel and Q355RE steel after RE alloying is analyzed for corrosion rate and Charpy V-notch impact energy. The focus is to understand the roles for RE on modifying inclusions and precipitates and to figure out the relationship between microstructure and properties.

2 Experimental

2.1 Specimens Design

A common commercial Q355 low alloy steel was chosen as the experimental material. The metal used for RE alloying was Ce and La mischmetal with the mass ratio of about 1:2. Two slabs in 230 mm thick were heated to 1240 °C for 5 h. After air-cooling to 1070 °C, followed by rolling via seven passes on a rolling mill, the slabs were rolled to plates of 35 mm thickness. The finish rolling was controlled at 880 °C and the final plates were rolled to 11.75 mm. Subsequently, the plate was hot-rolled at 660 °C and air cooled to room temperature. The chemical compositions of the two structural steels under investigation are listed in Table 1.

2.2 Test Procedure

Metallographic images and inclusion morphology were observed by optical microscopy (OM) and scanning electron microscopy (SEM) combined with energy dispersive X-ray spectroscopy (EDS). Inclusion identifying was processed by electrolytic extraction methods and SEM to observe the three-dimensional morphology directly. The schematic illustration of an electrolytic apparatus used for separating inclusions is show in Fig. 1. The electrolytic solution was composed of 30 g $FeSO_4$, 10 g NaCl and 20 g $C_6H_6Na_2O_7$ dissolved in 1000 mL distilled water. After electrolysis process, inclusions were obtained by centrifugal separation and fully dried to observe the morphology. ASPEX-SEM was used to detect on the area of 100 mm^2 randomly in the sample to characterize inclusions. Three samples were analyzed and average value of statistical results was taken in results.

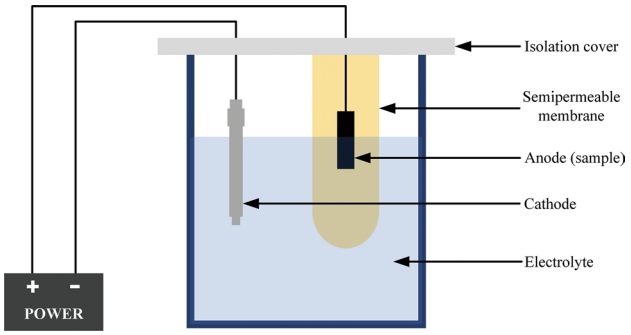


Fig. 1 Schematic illustration of an electrolytic apparatus used for separating inclusions

Three parallel samples (55 mm × 50 mm × 3 mm) were prepared for evaluating the rate of corrosion weight loss by the method of wet-dry immersion test. Before immersion test was undertaken, samples were cleaned by supersonic cleaner and weighed after fully dry. Then samples were exposed to a solution of 0.01 mol/L $NaHSO_3$ at 45 °C. The accelerated corrosion durations were determined at 12 h, 24 h, 48 h and 72 h according to the guideline of TB/T 1979–2014. The corrosion products on the surface of samples were removed by acid solution and the samples were weighed to determine the mass loss.

Charpy V-notch impact test was conducted with the samples of 10 mm × 10 mm × 55 mm. The samples were cut along the rolling direction, and the impact toughness data presented is an average of three measurements for each test condition. The transverse and longitudinal toughness were carried out at gradient temperature from − 60 to 0 °C. The fractograph and dislocation morphology were observed by SEM–EDS and transmission electron microscopy (TEM), respectively.

3 Results

3.1 Effects of RE Elements on Microstructure

Figure 2 depicts the optical micrographs of Q355 steel and Q355RE steel samples. The microstructure of Q355 steel exhibits a fine ferrite structure with minor granular bainite

Table 1 Chemical composition of experimental steels (mass%)

Sample	C	Si	Mn	P	S	Cr	Ni	Al	Cu	Ce	La
Q355 steel	0.15	0.21	1.26	0.014	0.007	0.024	0.013	0.049	0.021	–	–
Q355RE steel	0.15	0.22	1.25	0.014	0.007	0.023	0.013	0.049	0.021	0.0144	0.0064

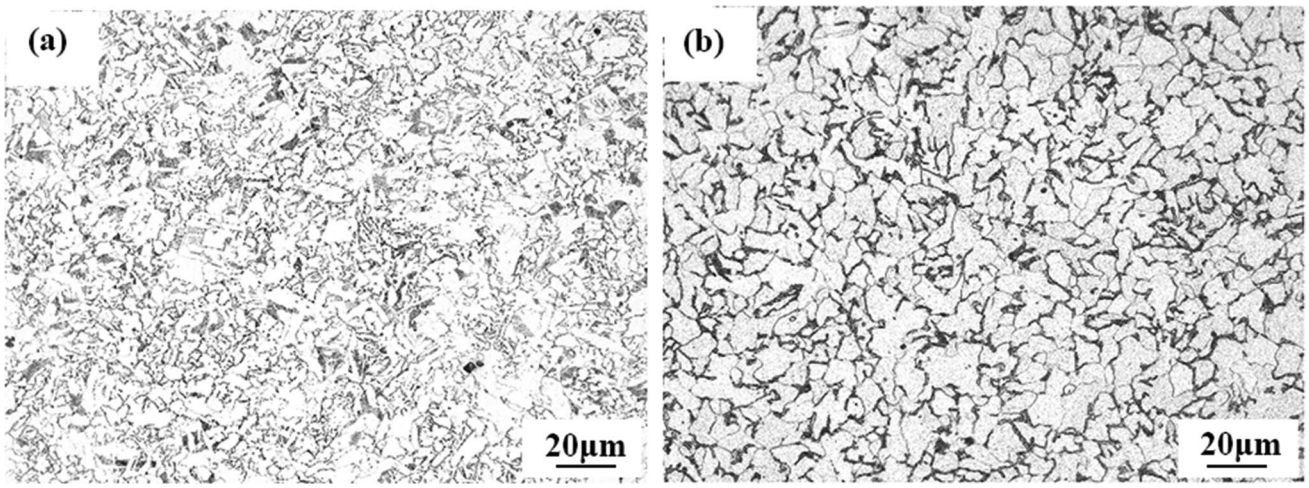


Fig. 2 Optical micrographs in **a** Q355 steel, **b** Q355RE steel

(Fig. 2a), while a mixed microstructure consisting of ferrite and pearlite is observed in Q355RE steel (Fig. 2b). The difference of these two microstructures indicates that the addition of RE elements has some influence on the transformation.

3.2 Effects of RE Elements on Inclusions

In order to determine the inclusion types, SEM observation and EDS analysis of inclusions are conducted, as shown in Fig. 3. Results shows that Al_2O_3 with irregular shape and

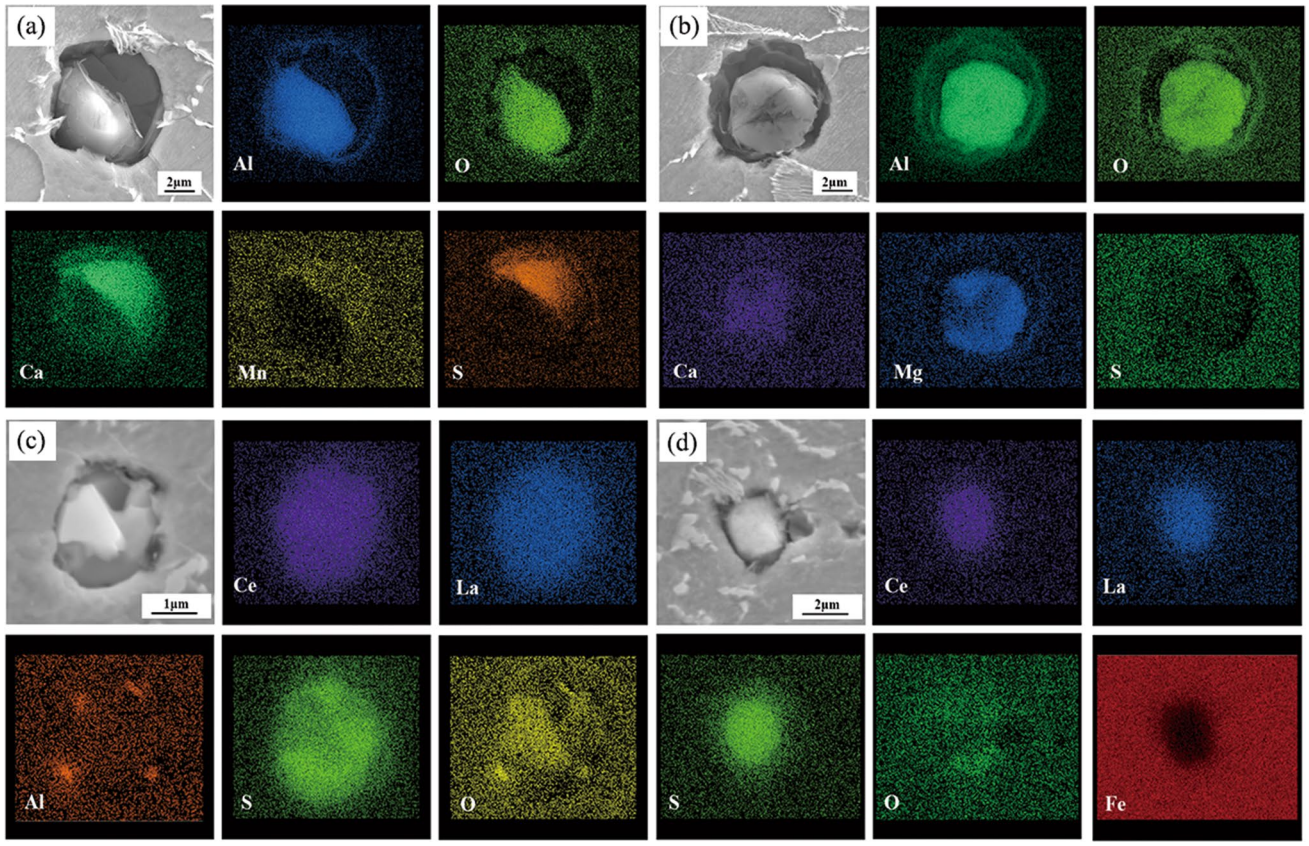


Fig. 3 SEM images and EDS mapping results of inclusions: **a** oxide, **b** complex inclusions in Q355 steel; and **c** RE oxy-sulfide, **d** RE sulfide in Q355RE steel

striplike MnS are the main inclusions formed in Q355 steel, and the enrichment of Mg and Ca in EDS mapping indicates the existence of CaO and MgO. After adding RE into Q355RE steel, Al_2O_3 and MnS are modified into spherical or ellipsoidal inclusions consisted of mainly RE sulfide and RE oxysulfide. Some RE complex inclusions may contain small amounts of Al_2O_3.

The three-dimensional morphology of inclusions by electrolytic extraction method is exhibited in Fig. 4. The inclusions in Q355 steel are mainly consisted of three types of inclusions, as shown in Fig. 4a. Small spherical inclusions with an average particle size of about 3 μm are recognized as CaS type inclusion (painted in green) by EDS attached to SEM. Irregular blocks with an average particle size of about 5 μm are determined as MnS type inclusion (painted in yellow), while inclusions with sharp angles are determined as Al_2O_3 type inclusion (painted in red). After RE alloying, inclusions are obviously refined and spheroidized (Fig. 4b). Almost all inclusions are modified into RE inclusions (painted in blue), except for a few Al_2O_3 inclusions.

3.3 Effects of RE Elements on Corrosion Resistance Properties

Figure 5 shows the relationship between corrosion rates and corrosion time obtained by wet-dry immersion tests. The corrosion rate ($g{\cdot}m^{-2}{\cdot}h^{-1}$) is obtained from the following known equation [21]:

$$W = \frac{G_0 - G_1}{S \times t}, \tag{1}$$

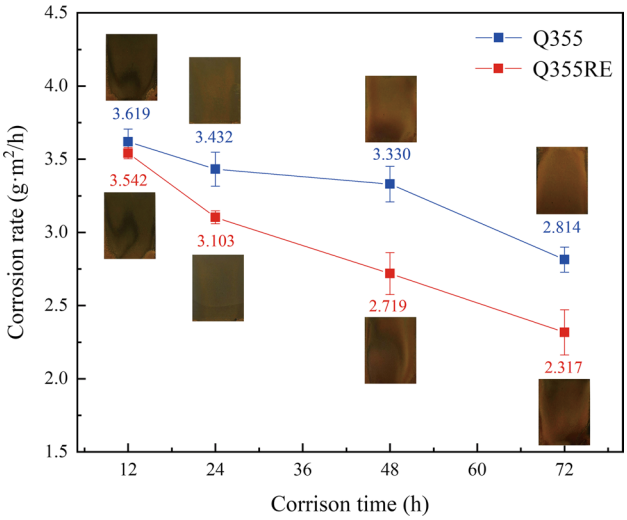


Fig. 5 Corrosion rates of Q355 steel and Q355RE steel obtained in wet-dry cycle immersion test

where W is corrosion rate ($g\ m^{-2}\ h^{-1}$), G_0 is initial sample weight before immersion test (g), G_1 is sample weight after immersion test (g), S is the initial total surface area of sample (m^2), t is corrosion time (h).

3.4 Charpy V-Notch Impact Properties

The variation of impact energy is measured to investigate the influence of RE, as described in Fig. 6. The absorbed impact energy decreases with a drop in temperature. The value of impact toughness of Q355RE samples is higher than that of Q355 samples. It is indicated that Q355RE steel displayed better impact fracture resistance than Q355 steel.

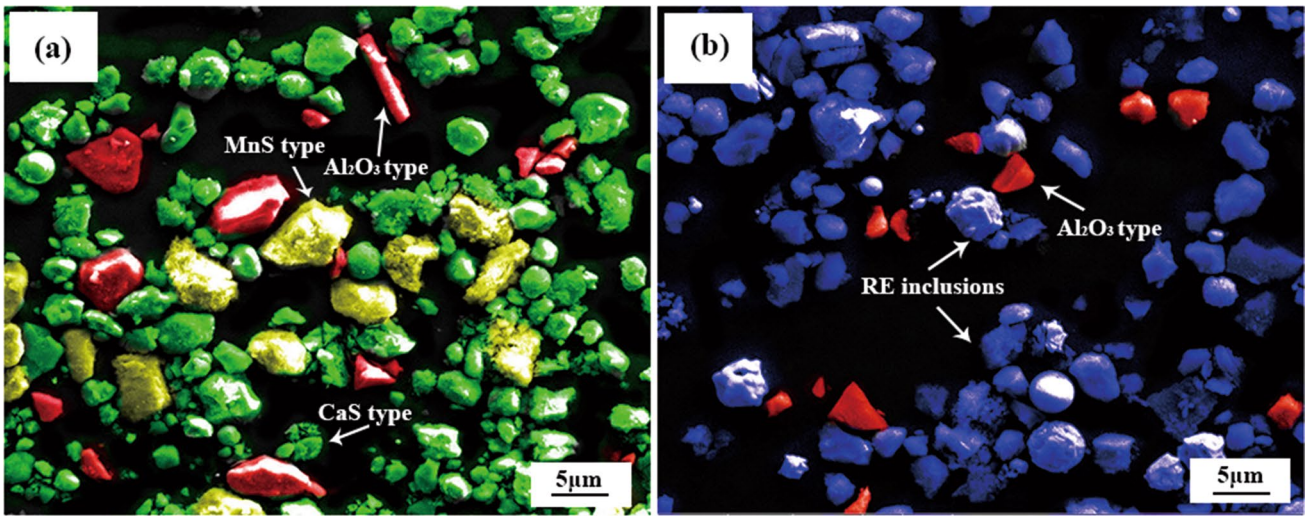


Fig. 4 Three-dimensional morphologies of inclusions (painted in different colors) obtained by the method of electrolytic extraction in **a** Q355 steel, **b** Q355RE steel

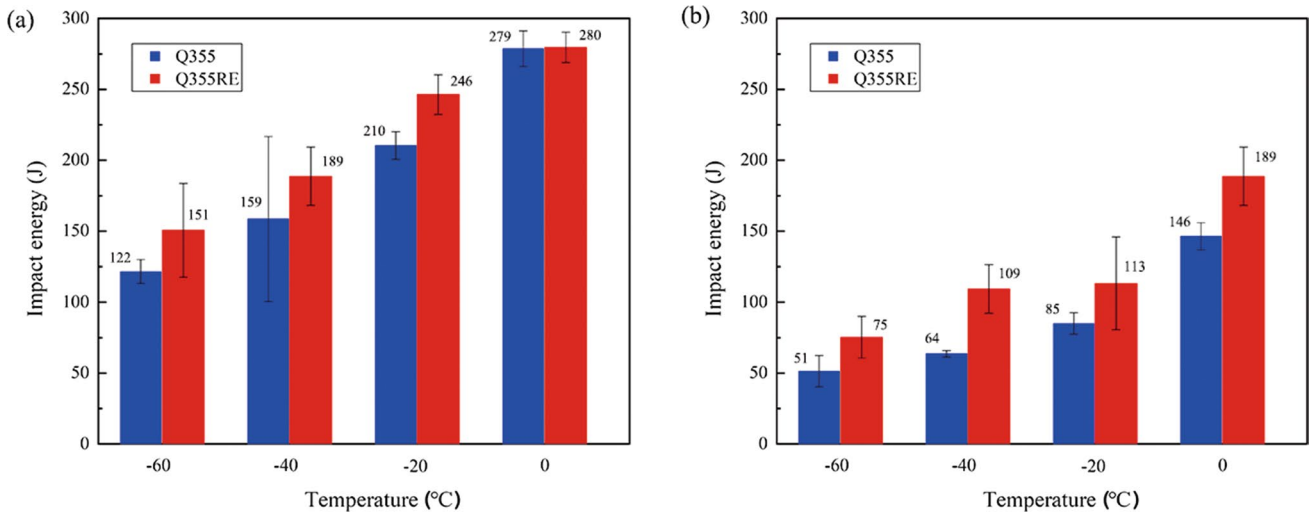


Fig. 6 Evolution of impact properties with temperature in **a** longitudinal direction, **b** transverse direction in Q355 steel and Q355RE steel

The improvement of steel toughness is more obvious in the transverse direction.

4 Discussion

4.1 Change of Microstructure

The change of microstructure proves that RE probably impedes the diffusion of carbon and solute atoms and increase the content of ferrite correspondingly, which is similar to the conclusion obtained by previous works [10, 22–24]. During the cooling process after hot rolling, the pre-eutectoid ferrite precipitates from austenite when cooling temperature reaches Ar3. When cooling temperature continues to drop below the temperature of Ar1, eutectoid decomposition occurs and cement (Fe_3C) appears. The formation of Fe_3C reduces the surrounding carbon concentration and promotes the nucleation of ferrite. Then ferrite continues to nucleate and grow on both sides of Fe_3C, and these two phases are integrated into pearlite in general. Due to the fast cooling rate, a small amount of bainite precipitates from the supersaturated ferrite matrix in the final state.

By comparing the metallographic structure of Q355 steel and Q355RE steel, it is obvious that RE elements can promote the formation of ferrite. During the growth of ferrite, carbon and other solute atoms concentrate into austenite and cause the transformation from retained austenite into pearlite or bainite [22]. When the temperature drops below transformation temperature of Bs, bainite is formed. However, the diffusion of RE as a solute atom is quite difficult due to its large atomic radius (Ce 0.1877 nm, La 0.1825 nm, Fe 0.1210 nm) [21]. The interaction coefficient between RE and carbon has been reported to be negatives [25].

4.2 Modification of Inclusions

RE element plays an important role on the shape control and modification of inclusions in Q355 steel, as shown in Figs. 3 and 4. The statistical results of various types of inclusions are listed in Table 2. A total number of 368 inclusions are detected. The proportion of sulfides including isolated MnS and complex sulfides is the highest, accounting for 59% of all detected inclusions. The other inclusions are mainly complex inclusions consisted of oxides (Al_2O_3) and sulfides (MnS and CaS). The inclusion size of MnS is the largest among all the inclusions with the average inclusion size of 4.7 μm. As a contrast, 362 inclusions were detected in the sample of Q355RE steel. The type of inclusions transfers from common oxides and sulfides to RE sulfide, RE oxide or RE complex inclusions. The amount of RE sulfide is the largest and accounts for 48.3% among all the inclusions with the average size of 2.7 μm. Due to the active chemical property of RE elements, it is concluded that RE elements tend to react with oxygen and sulfur firstly, which coincides with previous study focused on thermodynamic calculation results [19, 25].

By comparison, it is found that sulfide inclusions account for more than 50% in Q355 steel and the number of inclusions in Q355 steel is largely determined by the sulfur content of steel. The size distribution of inclusions is statistically evaluated and demonstrated in Fig. 7. It can be seen that the size of most inclusions in Q355 steel ranges from 1.5 to 3.5 μm, accounting for about 80% of all inclusions (Fig. 7a). The maximum size of inclusions is more than 10 μm, and inclusions with large size are mainly consisted of MnS and Al_2O_3 (Fig. 7b). After RE alloying, the maximum size of inclusion is less than 7.5 μm, and the average size of

Table 2 Type, counts and size distribution of inclusions in Q355 steel and Q355RE steel

Sample	Type of inclusions	< 1 μm	1–2 μm	2–3 μm	3–4 μm	4–5 μm	5–6 μm	> 6 μm	Total	Percentage (%)	Average size (μm)	Maximum size (μm)
Q355 steel	CaS-MnS	0	57	59	17	5	4	3	145	39.4	2.4	7.3
	MnS	0	28	17	10	4	6	7	72	19.6	4.7	7.8
	CaS-Al_2O_3-MnS	0	5	71	31	6	4	5	122	33.1	3.8	5.4
	Al_2O_3-MnS	0	0	2	7	9	3	8	29	7.9	4.5	10.3
Q355RE steel	RES	0	48	76	31	14	5	1	175	48.3	2.7	7.4
	RE_2O_3	0	32	61	10	2	2	0	107	29.6	2.3	5.7
	RES-RE_2O_3	0	17	19	14	1	0	0	51	14.1	2.8	5.4
	RES-RE_2O_3-Al_2O_3	0	6	10	12	1	0	0	29	8.0	3.0	5.3

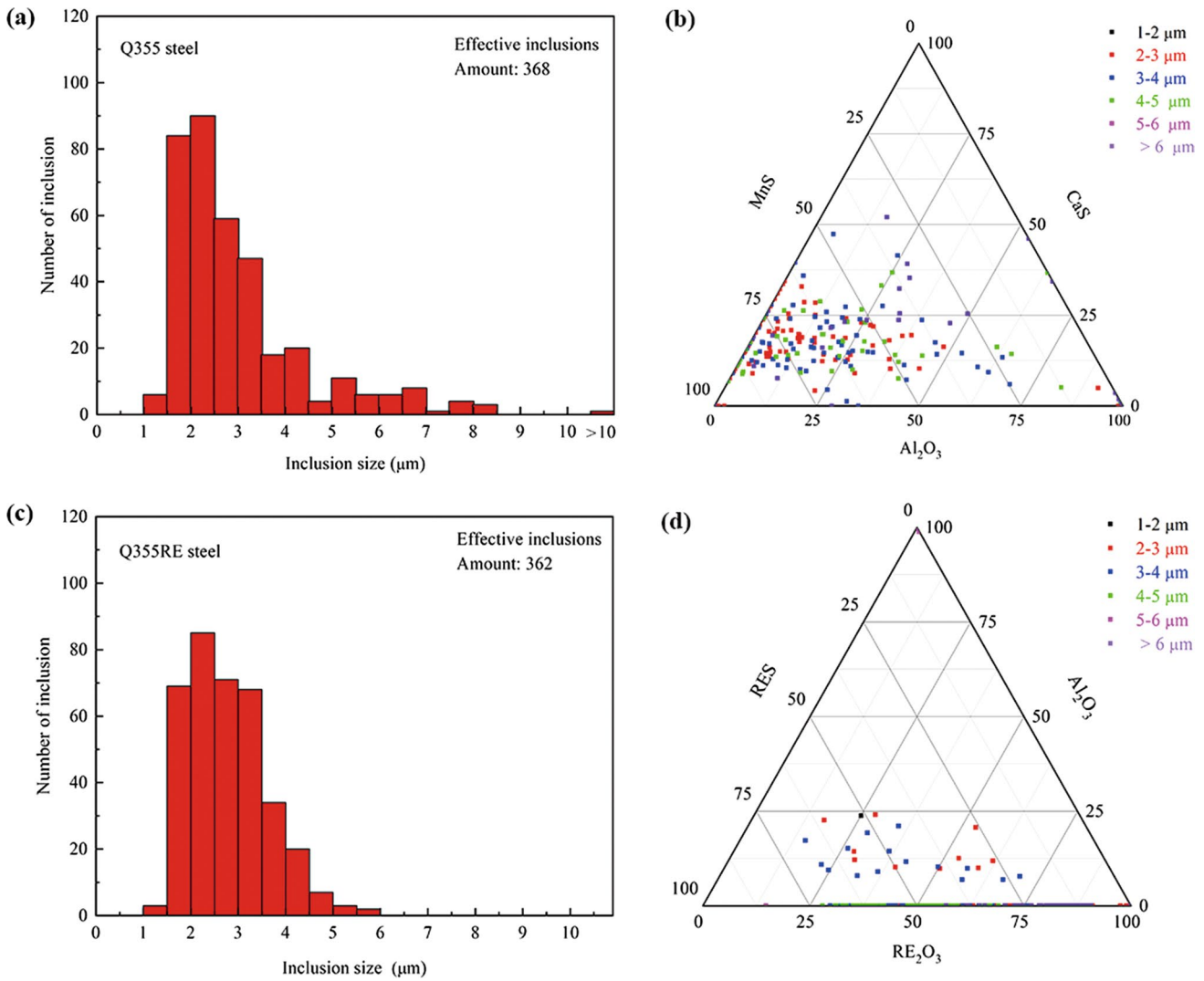


Fig. 7 Size distribution and ternary phase diagram of inclusions in **a**, **b** Q355 steel, **c**, **d** Q355RE steel

inclusions has been decreased obviously (Fig. 7c). Special attention should be paid to the detection of these impurities in steels. As is known to all, cracks are easy to initiate on inclusion particles that are with large size, irregular sharp and high hardness [13, 26]. It is apparent that the addition of RE can make the inclusions smaller and more dispersed,

which is beneficial for enhancing comprehensive properties of test steel.

4.3 Mechanism of Corrosion Resistance Improvement

Corrosion rate is closely related to the formation of rust layers on sample surface. The morphology of oxide products on the surface of rust layers is demonstrated in Fig. 8. The circular corrosion morphology is formed on the surface of samples under cycled wet-dry immersion tests, which is due to the deposition of liquid on the surface unevenly. It is obvious that the corrosion uniformity of Q355RE steel is much better than that of Q355 steel. The rust layer particles turn finer in Q355RE steel after RE alloying. A small amount of oxide particles containing RE is found on the surface of rust layer, as shown in the red circle in Fig. 8d. RE oxide particles also contain other alloying elements such as Al and Si, indicating that RE inclusions participate in the corrosion reaction of Q355RE steel.

By analyzing the XRD patterns of corrosion products, the main phases consist of γ-FeOOH, α-FeOOH and Fe_3O_4/γ-Fe_2O_3 (Fig. 9). Based on previous study, it is known that unstable γ-FeOOH is an electrochemical active phase and is formed in the initial stage of corrosion, which can transform into Fe_3O_4/γ-Fe_2O_3 and α-FeOOH during the corrosion process [27, 28]. Stable α-FeOOH is the most protective hydroxy iron oxide and prevents corrosive ions from invading the matrix. Fe_3O_4 and γ-Fe_2O_3 have the similar XRD patterns and it is quite difficult to distinguish accurately whether it is a mixture or a single compound.

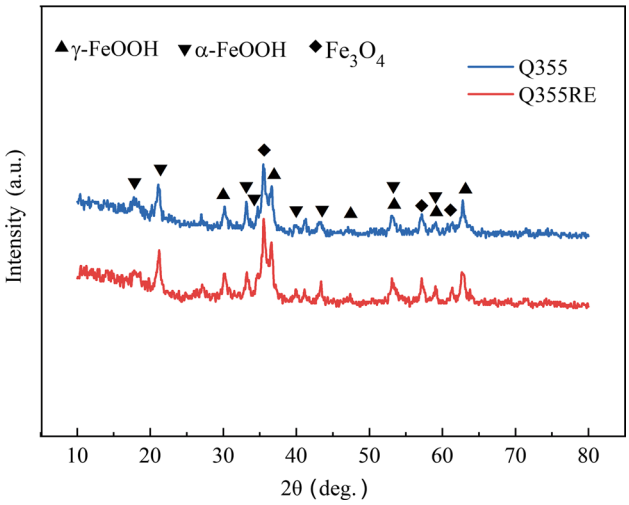


Fig. 9 XRD patterns of oxide products of Q355 steel and Q355RE steel

Table 3 shows the relative content of different corrosion products. The content of stable α-FeOOH is higher and the content of unstable γ-FeOOH is lower in Q355RE steel compared to those in Q355 steel. The content of Fe_3O_4/γ-Fe_2O_3 is decreased slightly. The ratio of α-FeOOH/γ-FeOOH (α/γ)

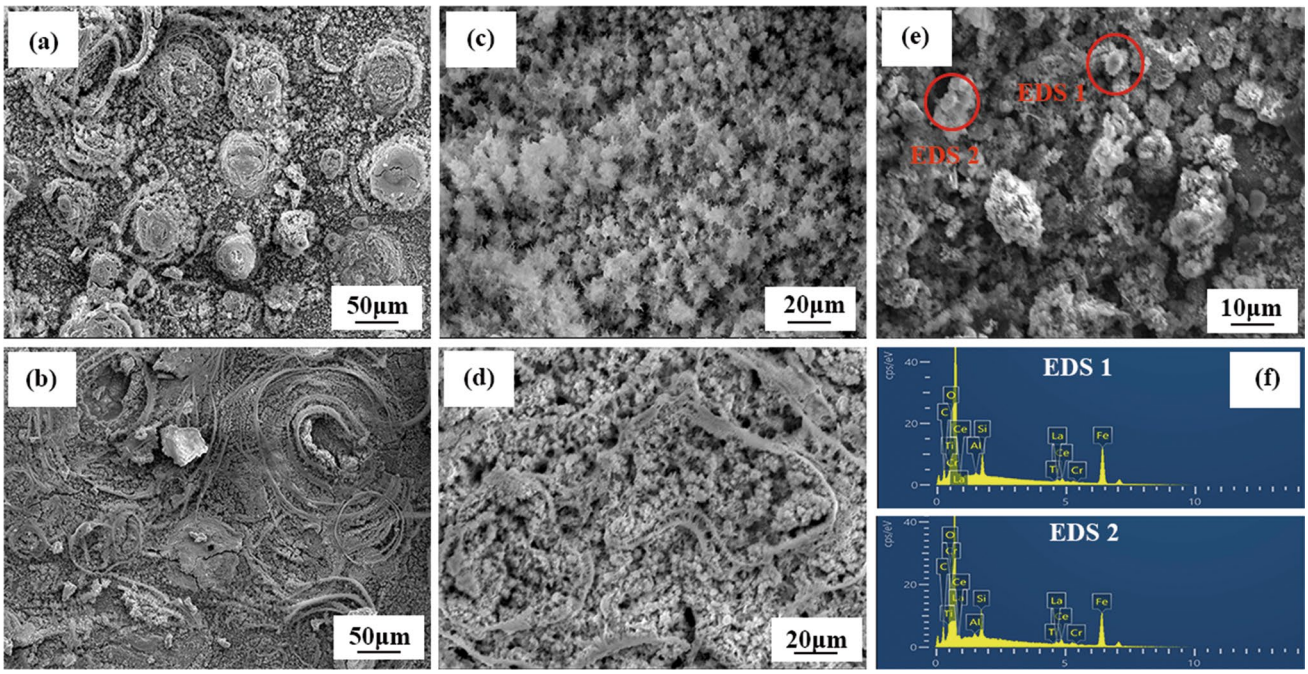


Fig. 8 SEM surface morphologies of rust layers in **a**, **c** Q355 steel; **b**, **d** Q355RE steel; **e**, **f** RE oxide particles and its corresponding EDS patterns

Table 3 Relative content of corrosion products by XRD analysis (mass%)

Sample	α-FeOOH	Fe_3O_4/γ-Fe_2O_3	γ-FeOOH	α/γ
Q355 steel	40.8	55.8	2.1	19.4
Q355RE steel	46.2	54.3	1.5	35.5

is an important parameter to evaluate whether corrosion products are stable [29]. From the obtained date in Table 3, the value of α/γ increases from 19.4 to 35.5, which indicates that RE alloying improves the stability of rust layers and the ability of protecting the matrix, thus delaying the corrosion. It has been reported in our previous study that S^{2-} and HS^- produced by dissolved MnS can promote local corrosion, including further occurrence and expansion of pitting in plain low carbon steels [21]. After adding RE element into Q355 steel, RE can modify inclusions into small-sized spherical inclusions to reduce the size of phase boundary between inclusions and matrix. It effectively reduces the active area at the surface of steel and weakens micro-region electrochemical corrosion. RE inclusions have also exhibited a much lower Volta potential than the matrix. Furthermore, some oxide particles containing RE is found on the surface of rust layer, indicating that RE inclusions participate in the corrosion reaction of Q355RE steel. Therefore, RE inclusions dissolve and precipitate during corrosion process and part of them exist in the form of insoluble RE compounds in the rust layers.

4.4 Mechanism of Impact Property Improvement

The fracture surfaces of impact specimens in the transverse direction are shown in Fig. 10. The observation indicated that a clear transition from ductile to brittle behavior with a drop of temperature, and the change of predominant mechanism of fracture has been shown evidentially. Fine and deep dimples for Q355 steel and larger ones for Q355RE steel

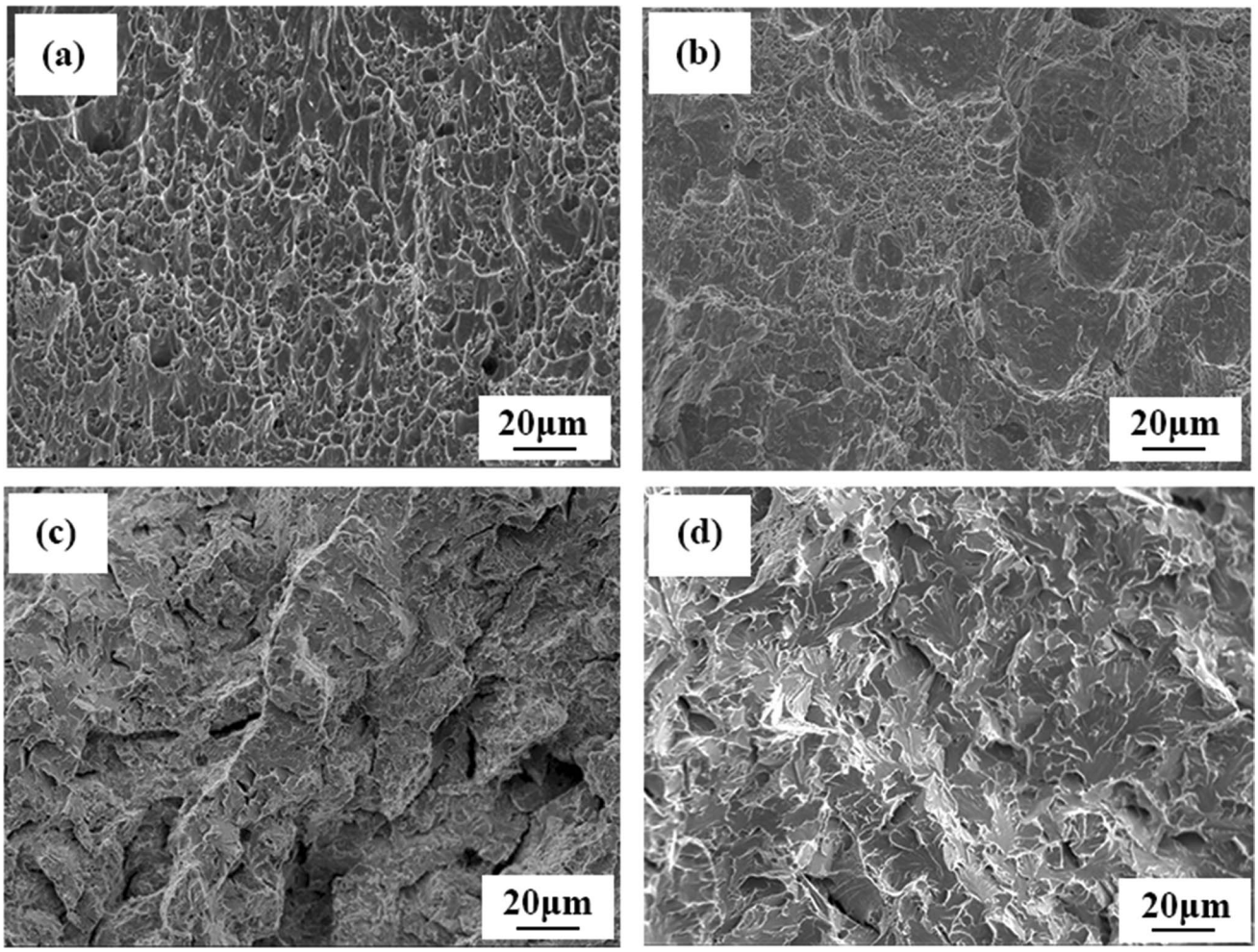


Fig. 10 SEM fractographs of investigated fractured steels in Q355 steel at the temperature of **a** 0 °C, **c** − 60 °C; in Q355RE steel at the temperature of **b** 0 °C, **d** − 60 °C

under the test temperature of 0 °C are observed in Fig. 10a and b. In contrast, the fracture morphology displayed brittle fracture when impact test temperature decreased to − 60 °C, as shown in Fig. 10c and d. The formation of cleavage facets has been the main fracture mechanism. It should be noted that more secondary cracks exist on the surface of fractured samples in Q355 steel than that in Q355RE steel, which indicates that the formation and propagation of crack sources are inhibited by RE alloying.

Impact fracture analysis of transverse samples is performed to display the morphology of impact fracture, as shown in Fig. 11. Impact properties are a combination evaluation of strength and plasticity and notched impact samples are more sensitive to the variation of microstructure [30, 31]. The main reason for increasing the impact energy after RE alloying can be attributed to the modification of inclusions. Angular inclusions may lead to micro-cracks initiation under loading and large-sized inclusions are detrimental to mechanical properties due to the difference of hardness and thermal expansion properties between matrix and inclusions. It is noted that the presence of inclusion (MnS, Al_2O_3) act as potential sites for crack initiation [32, 33]. It demonstrates that plastic deformation occurs in matrix and cracks begin to initiate and propagate at both ends of large inclusions when the load is applied in Q355 steel. Then the internal and boundary cracks occur around inclusion. Ultimately, the inclusions are separated from the matrix and the whole inclusion is regarded as the crack source, as shown in Fig. 11a. The larger the size of inclusions, the faster the rate of crack propagation after plastic deformation. When RE elements is added into steels, sulfur and RE elements act to form sub-micro spherical RE sulfides and composite inclusions ($Al_2O_3 + MnS + CaS$), which replaces striplike MnS. The shape of RE inclusions is mostly spherical or ellipsoidal with good morphology and distribution. Compared to sulfide inclusions, the linear expansion coefficient and elastic modulus of RE inclusions are similar to those in the matrix, which can effectively reduce the stress concentration around inclusions during impact test [34]. Small spherical RE inclusions have good cohesion with matrix and are less likely to be the crack sources, as indicated in Fig. 11b. It has been reported that RE inclusions with certain size range can produce large plastic deformation during the impact test and can alter the path of crack growth [34, 35]. Simultaneously, the activity of

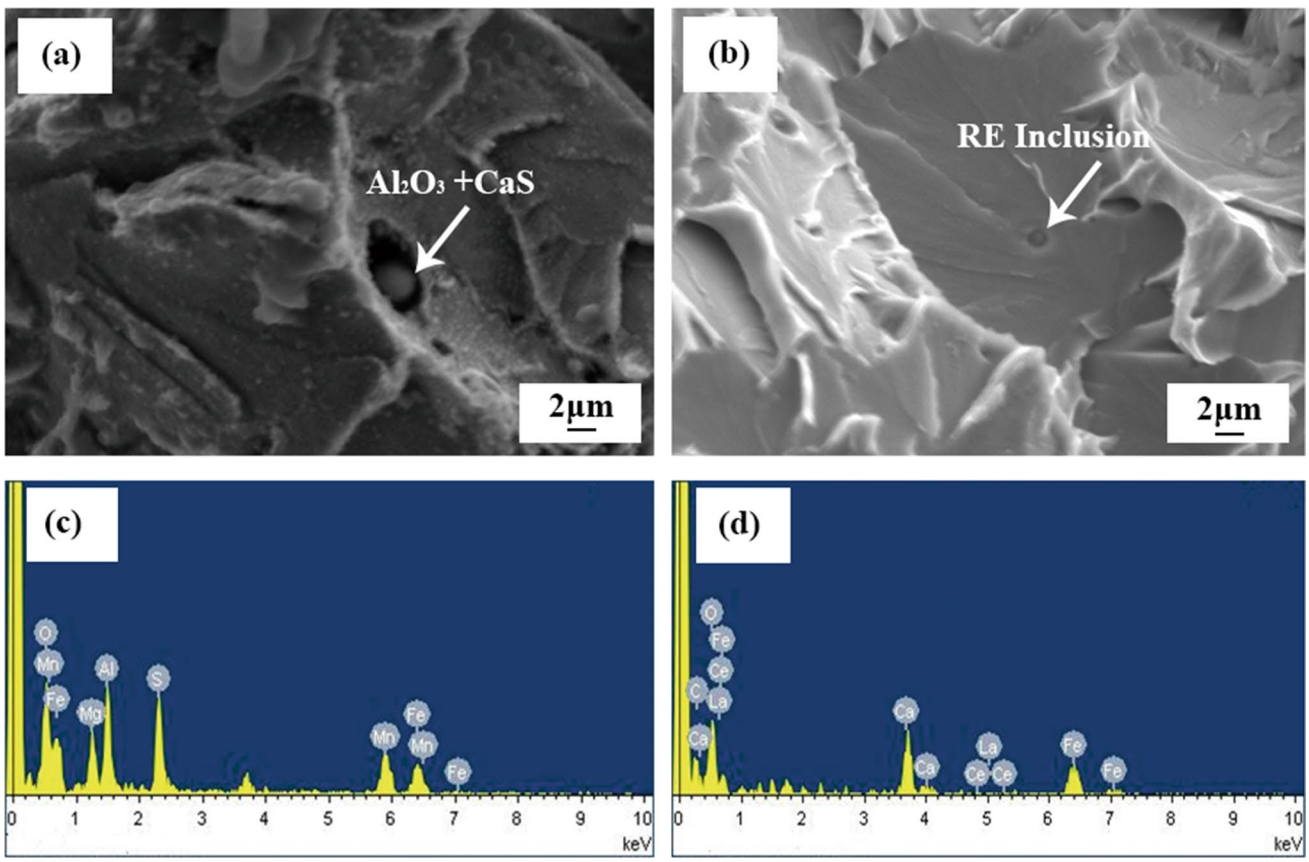


Fig.11 SEM morphologies of crack initiation and EDS analysis of inclusions in the fractured samples at temperature of − 60 °C in **a**, **c** Q355 steel; **b**, **d** Q355RE steel

small-sized RE inclusions is calculated to be lower than that of other large-sized inclusions in crack initiation and propagation [21, 36].

Figure 12 manifests the microstructure of the fractured samples in Q355RE steel by TEM. It is found that high-density dislocations and precipitates exist in the sample. The morphology results clearly demonstrate the precipitation of second phases along grain boundaries and its corresponding EDS mapping result shows that the spherical nano-scale precipitate is enriched with Ce, La, O, S and P (Fig. 12c). The nano-scale precipitates almost have little detrimental effect on mechanical properties investigated in

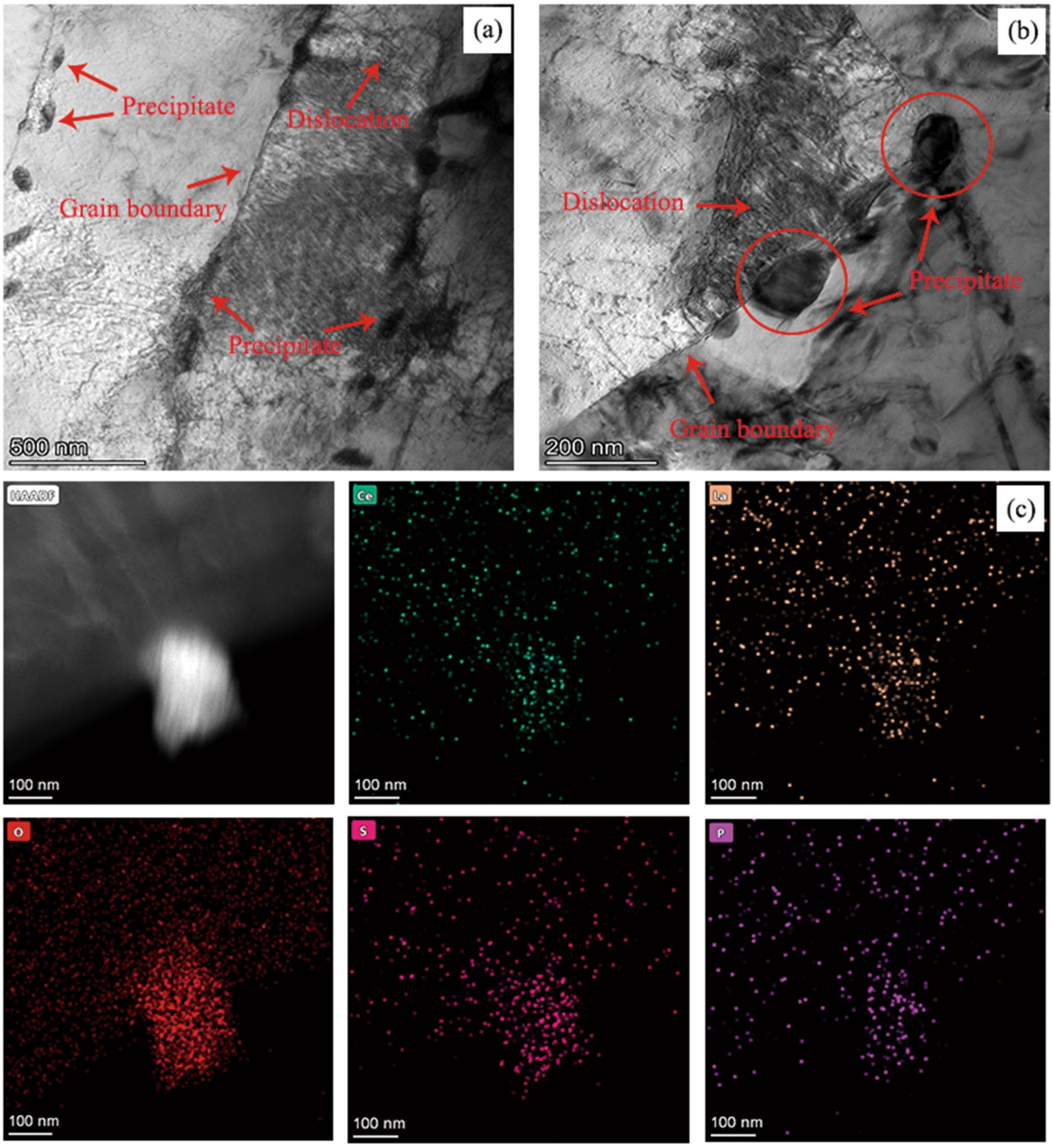


Fig.12 **a**, **b** Bright-field TEM images of dislocations and precipitates, **c** scanning tunneling electron microscopy, high-angle annular dark field (STEM-HAADF) image and mapping of nano-scale precipitate in Q355RE steel

many studies [37, 38]. Additionally, the precipitate purifies the grain boundaries. S is considered to be the most detrimental impurity element due to its tendency to segregate towards grain boundaries and the detrimental effect of S on the mechanical properties has been investigated. According to the theory of grain boundary segregation, the impurity elements such as S, P and Sb segregate towards grain boundaries, which weaken grain boundaries and leads to brittle fracture [39, 40]. RE has been reported to the depressing deleterious effects of low melting point metals in the way of fixation of tramp elements and minimizing amount of Fe–P phase [18]. This could be provided as a further evidence that stress concentration caused by element segregation is eliminated by purification of grain boundaries, leading to the enhancement of impact toughness. S can also gather at grain boundaries to inhibit uniform corrosion rate and is more likely to cause pitting corrosion [41]. Grain boundary purification is helpful to improve both impact toughness and corrosion properties.

Generally, nano-scale precipitate combines well with the matrix, and no split between precipitate and matrix occurs during impact test. Furthermore, the segregation of RE at grain boundaries and the interactions with other elements should be considered. More works need to further study whether a direct correlation between nano-scale precipitate and mechanical properties.

5 Conclusions

1. The microstructure of Q355RE steel was consisted of ferrite and pearlite and that of Q355 steel consisted of ferrite and bainite, which was due to the effect of RE on transformation temperature. Small-sized spherical RE inclusions (RES, RE_2O_3) replaced elongated MnS and large-sized Al_2O_3 in Q355RE steel after modifying by RE elements.
2. Compared to Q355 steel, Q355RE steel had lower corrosion rate. The value of α/γ was increased after RE alloying, indicating that more stable oxide products were formed on the surface of Q355RE steel. The decrease in corrosion rate was closely related to the formation of stable and dense rust layers promoted by RE alloying.
3. Q355RE steel also exhibited better resistance to fracture at low temperature, owing to the presence of RE modification to inclusions and its effects on reducing crack initiation.
4. RE precipitates containing sulfur and phosphorus was observed along grain boundaries. The purification of grain boundaries by RE was probably beneficial to the improvement of corrosion and impact properties.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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